§25.2 Definition and Expansions

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§25.2(i) Definition

When $\Re s > 1$,

25.2.1 $$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Elsewhere $\zeta(s)$ is defined by analytic continuation. It is a meromorphic function whose only singularity in $\mathbb{C}$ is a simple pole at $s = 1$, with residue 1.

§25.2(ii) Other Infinite Series

25.2.2 $$\zeta(s) = \frac{1}{1-2^{-s}} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^s},$$

$\Re s > 1$.

25.2.3 $$\zeta(s) = \frac{1}{1-2^{1-s}} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^s},$$

$\Re s > 0$.

25.2.4 $$\zeta(s) = \frac{1}{s-1} + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \gamma_n (s-1)^n,$$

where the Stieltjes constants γ_n are defined via

25.2.5 $$\gamma_n = \lim_{m\to\infty} \left(\sum_{k=1}^{m} \frac{(\ln k)^n}{k} - \frac{(\ln m)^{n+1}}{n+1} \right).$$

25.2.6 $$\zeta'(s) = -\sum_{n=2}^{\infty} (\ln n) n^{-s},$$

$\Re s > 1$.

25.2.7 $$\zeta^{(k)}(s) = (-1)^k \sum_{n=2}^{\infty} (\ln n)^k n^{-s},$$

$\Re s > 1$, $k = 1, 2, 3, \ldots$.

For further expansions of functions similar to (25.2.1) (Dirichlet series) see §27.4. This includes, for example, $1/\zeta(s)$.

§25.2(iii) Representations by the Euler–Maclaurin Formula

25.2.8 $$\zeta(s) = \sum_{k=1}^{N} \frac{1}{k^s} + \frac{N^{1-s}}{s-1} - s\int_N^{\infty} \frac{x - \lfloor x \rfloor}{x^{s+1}}\,dx,$$

$\Re s > 0$, $N = 1, 2, 3, \ldots$.

25.2.9 $$\zeta(s) = \sum_{k=1}^{N} \frac{1}{k^s} + \frac{N^{1-s}}{s-1} - \frac{1}{2}N^{-s} + \sum_{k=1}^{n} \binom{s+2k-2}{2k-1} \frac{B_{2k}}{2k} N^{1-s-2k} - \binom{s+2n}{2n+1} \int_N^{\infty} \frac{\widetilde{B}_{2n+1}(x)}{x^{s+2n+1}}\,dx,$$

$\Re s > -2n$; $n, N = 1, 2, 3, \ldots$.

25.2.10 $$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} + \sum_{k=1}^{n} \binom{s+2k-2}{2k-1} \frac{B_{2k}}{2k} - \binom{s+2n}{2n+1} \int_1^{\infty} \frac{\widetilde{B}_{2n+1}(x)}{x^{s+2n+1}}\,dx,$$

$\Re s > -2n$, $n = 1, 2, 3, \ldots$.

For B_{2k} see §24.2(i), and for $\widetilde{B}_n(x)$ see §24.2(iii).

§25.2(iv) Infinite Products

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25.2.11 $$\zeta(s) = \prod_{p}(1 - p^{-s})^{-1},$$

$$\Re s > 1,$$

product over all primes p.

25.2.12 $$\zeta(s) = \frac{(2\pi)^{s}\mathrm{e}^{-s-(\gamma s/2)}}{2(s-1)\Gamma\left(\frac{1}{2}s+1\right)}\prod_{\rho}\left(1-\frac{s}{\rho}\right)\mathrm{e}^{s/\rho},$$

product over zeros ρ of ζ with $\Re\rho > 0$ (see §25.10(i)); γ is Euler's constant (§5.2(ii)).

25.1 Special Notation 25.3 Graphics

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